

Reactivities of mixed organozinc and mixed organocopper reagents. Part 13[†] Kinetic study for phosphine-catalyzed acylation of alkylarylzincs and effect of residual group on the transfer rate of alkyl group[‡]

Özgen Ömür Pekel^{a*}

Kinetics of reactions of di-*n*-butylzinc, *n*-Bu₂Zn, and mixed *n*-butyl(substituted phenyl)zinc reagents and *n*-Bu(functional group (FG)-C₆H₄)Zn with benzoyl chloride in the presence of tri-*n*-butylphosphine have been investigated. Reaction rates of transferable *n*-butyl group have been determined in tetrahydrofuran at 0 °C to compare the transfer rate of *n*-butyl group in homo and mixed diorganozincs. Rate law is consistent with a third-order reaction, which is first order in diorganozinc, benzoyl chloride, and *n*-Bu₃P, and a mechanism was proposed. The lower reaction rate of *n*-BuPhZn than that of *n*-Bu₂Zn and negative reaction constant in Hammett plot are in accordance with the carbanionic charge of transferable *n*-butyl group in the acylation reaction. These findings support the hypothesis that the reaction rate of transferable group, R_T, changes depending upon the residual group, R_R, in R_RR_TZn reagents. Copyright © 2016 John Wiley & Sons, Ltd.

Keywords: acylation of diorganozincs; kinetics of diorganozincs; mixed diorganozincs; phosphine catalyst

INTRODUCTION

Diorganozinc reagents, R₂Zn, and triorganozincates, R₃ZnM (M=Li, Na, MgX), which are most reactive than RZnCl, are one of the most used organometallic reagents due to their functional group compatibility to many functional groups.^[1–3] Diorganocuprates, R₂CuM (M=Li, MgX, ZnX; X=Br, Cl), are also commonly used in organic synthesis due to their soft nucleophilicity toward carbon electrophiles.^[4–6] However, in the reactions of R₂CuM, R₂Zn, and R₃ZnM reagents, a synthetic problem arises because only one of the R groups can be transferred to the electrophile. This limitation to the atom-economic use of R₂CuM, R₂Zn, and R₃ZnM reagents has been solved by using either (i) mixed organocopper and organozinc reagents of R¹R²CuM,^[7–16] R¹R²Zn,^[17–31] and (R¹)₂R²ZnM^[32–37] type, in which the R² group has a lower transfer rate than R¹ group, or (ii) mixed organocopper and organozinc reagents of R_RR_TCuM,^[7,8,38–46] R_RR_TZn,^[47–55] and (R_R)₂R_TZnM type,^[7,38,56–68] in which R_R is residual (dummy) group and R_T is transferable group.

In the mixed diorganocuprates, R_RR_TCuM, both organyl groups^[38–41] and nonorganyl groups, such as RS, R₂N, and R₂P,^[42–45] have been frequently used as residual groups, R_R. The group selectivity in the reactions of R¹R²CuLi reagents was found to be in the order of *n*-Bu ~ *s*-Bu > *t*-Bu; Ph > alkenyl > Me > alkynyl in their C–C coupling reactions,^[7,9,10] however, in the reversed order of Me > alkynyl > alkenyl in their 1,4-addition reactions.^[11] In the reactions of (FG–C₆H₄)RCuLi, in which R is Me₂CPhCH₂ or Me₃CCH₂, transfer of functionalized aryl group takes place selectively.^[8]

Mixed diorganozinc reagents of both R¹R²Zn and R_RR_TZn types have been greatly used in the 1,2-addition^[47–50] and 1,4-addition

reactions^[51–54] and group selectivity in reactions of R¹R²Zn reagents is in the order of alkenyl ~ phenyl > Me > Et > *t*-Bu.

In their systematic work,^[56–66] Erdik and coworkers observed that the group selectivities of mixed diorganocuprates, R¹R²CuM (M=MgBr),^[56–58] ZnI,^[59] diorganozincs, R¹R²Zn,^[58,60–66] and Cu-catalyzed mixed triorganozincates, R¹(R²)₂ZnMgBr,^[58,63] depend on the reaction parameters, that is, the electrophilic substrate, the solvent, and the temperature as well as the transition metal catalyst and organocatalyst, and reported that the group selectivity of mixed organocopper and organozinc reagents can be controlled by changing the reaction parameters. They also developed new atom-economic synthetic procedures for C–C,^[56–58,63,64,66] C-acyl,^[58–61] and C–N^[62] coupling employing mixed arylzinc reagents. However, probing the origin of group selectivity in the mixed organocopper and organozinc reagents still seems a problem.

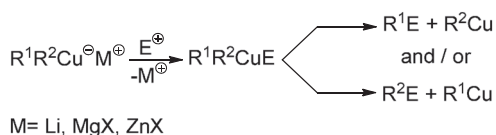
The group selectivity of mixed cuprates, R¹R²CuLi, is considered to be a function of R¹E or R²E formation in the reductive elimination of Cu(III) intermediates, R¹R²CuE, which is formed in the oxidative addition step (Scheme 1). The long accepted hypothesis in

* Correspondence to: Ö. Ömür Pekel, Science Faculty, Ankara University, Beşevler, Ankara 06100, Turkey.
E-mail: oomur@science.ankara.edu.tr

[†] The number of the serial work has been used with the permission of Ender Erdik.

[‡] This paper is dedicated to my PhD Supervisor Prof. Dr Ender Erdik.

a Ö. Ömür Pekel
Ankara University, Science Faculty, Beşevler, Ankara 06100, Turkey



Scheme 1. Group selectivity in the reaction of mixed diorganocuprates with an electrophile

substitution and 1,4-addition of mixed diorganocuprates is that the group with a stronger Cu–C bond acts as the group of lower selectivity, that is, a better residual group.^[7,38,67,68] Theoretical studies of Nakamura established that the group selectivity is controlled by thermodynamic stability and kinetic reactivity in the reductive elimination of triorganocopper(III) intermediates formed in these reactions.^[69,70]

Bertz reported that R_R groups in R_RR_TCuLi reagents impart peculiar properties to π -complexes of the cuprates formed in 1,4-addition reactions.^[71] Recently, Nakamura introduced the intrinsic effects of organyl groups on the reactivity and selectivity of gas phase alkylation reactions between mixed methyl cuprates, $[RMeCu]^{\ominus}$ (R = Alkyl, benzyl, phenyl), and methyl iodide.^[72]

Erdik has also hypothesized that the reaction rate of R^2 group in mixed diorganocuprates, R^1R^2CuM (M = Li, MgX, ZnX), and diorganozincs, R^1R^2Zn , should depend on R^1 group leading to a rate different from those of homo diorganocuprates, $(R^2)_2CuM$, and homo diorganozincs, $(R^2)_2Zn$, respectively. A series of kinetic studies on the reactions of mixed diorganocuprates and diorganozincs supported this hypothesis. They observed that the alkylation rate of transferable n -Bu group in n -BuPhCuMgBr is lower than that in n -Bu₂CuMgBr^[57] and also acylation rate of n -Bu group in n -BuPhCuZnI is lower than that in n -Bu₂CuZnI.^[59] In addition, alkylation rate of n -Bu group in n -Bu(FG–C₆H₄)CuMgBr reagents and acylation rate of n -Bu group in n -Bu(FG–C₆H₄)CuZnI reagents change depending on FG substituents. However, alkylation rate of Ph group in catalytic n -BuPhCuMgBr, which is derived from n -BuPh₂ZnMgBr and CuI catalyst, was found higher than that of catalytic homocuprate Ph₂CuMgBr.^[63] They also compared the reaction rate of transferable Ph group in allylation of (n -alkyl)PhZn reagents and found that it changes depending on n -alkyl group, and all higher than that in Ph₂Zn.^[65]

In connection with my recent studies on the control of group selectivity in acylation of n -alkylphenylzinc reagents^[60,61] and kinetics of acylation of iodozinc alkylarylcuprates,^[64] I focused my interest on the acylation of n -alkylarylzinc reagents with the objectives of kinetics and mechanism of phosphine-catalyzed acylation of diorganozincs and providing another kinetic support for Erdik's hypothesis regarding the dependence of R_T group transfer ability on R_R group in reactions of R_RR_TZn reagents. According to the best of my knowledge, there is a limited number of reported work on the kinetics of reactions of uncatalyzed^[60,61] and Cu(I)- and Ni(II)-catalyzed reactions^[56,62] of monoorganozincs and diorganozincs. A theoretical study for transfer ability in mixed diorganozincs has been also published.^[71] Herein, I report a detailed kinetic investigation of acylation of n -alkyl(FG–C₆H₄)Zn reagents with benzoyl chloride and discuss the effect of residual FG–C₆H₄ group on the transfer rate n -alkyl group in acylation.

EXPERIMENTAL

All reactions were carried out under dry nitrogen atmosphere using oven-dried glassware. Reagents and solvents were handled

by using standard syringe–septum cap techniques.^[73] Quantitative gas–liquid chromatography analysis was performed on a Thermo Finnigan (San Jose, CA, USA) gas chromatograph equipped with a ZB-5 capillary column packed with phenylpolysiloxane using internal standard technique. Tetrahydrofuran (THF) was distilled from sodium benzophenone dianion. n -Butyl bromide, bromobenzene, and 4-methyl-, 3-methyl-, 4-methoxy-, 3-methoxy-, and 4-bromobenzenes were obtained commercially (Sigma-Aldrich, St. Louis, MO, USA) and were used without further purification. Mg turning for Grignard reagents was used without further purification. ZnCl₂ (Merck, Kenilworth, NJ, USA) was dried under reduced pressure at 100 °C for 2 h and used as a THF solution. Benzoyl chloride (Sigma-Aldrich) and n -Bu₃P (Sigma-Aldrich) were obtained commercially and were used without further purification. Grignard reagents n -BuMgBr and FG–C₆H₄MgBr (FG = H **b**, 4-Me **c**, 3-Me **d**, 4-MeO **e**, 3-MeO **f**, 4-Br **g**) were prepared in THF by standard methods and their concentration were determined by titration prior to use.^[74]

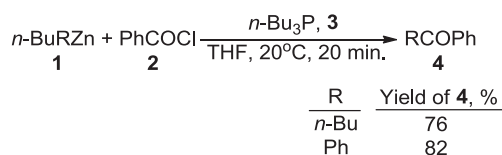
For the preparation of n -butyl(aryl)zinc reagents n -Bu(FG–C₆H₄)Zn (**1ab–1ag**) (FG = H **b**, 4-Me **c**, 3-Me **d**, 4-MeO **e**, 3-MeO **f**, 4-Br **g**),^[59] arylzinc chlorides FG–C₆H₄ZnCl were reacted with n -BuMgBr. FG–C₆H₄ZnCl were prepared by addition of arylmagnesium bromide (10 mmol) to ZnCl₂ (10 mmol) in THF (10 ml) at –20 °C with stirring at that temperature for 15 min. To freshly prepare FG–C₆H₄ZnCl reagent (10 mmol), n -BuMgBr (10 mmol) in THF was added dropwise and the mixture was stirred at –20 °C for another 15 min.

Homo di- n -butylzinc n -Bu₂Zn (**1a₂**) reagent was prepared by addition of 2 mol equivalent of n -BuMgBr reagent in THF to a solution of 1 mol equivalent of ZnCl₂ in THF at –20 °C, stirring at that temperature for 20 min.

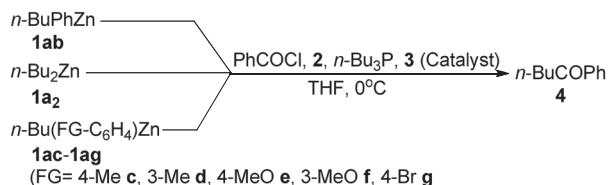
A typical kinetic procedure for reaction of n -BuPhZn (**1ab**) with benzoyl chloride **2** catalyzed by n -Bu₃P in THF at 0 °C under pseudo-first-order conditions in the presence of excess **1ab** is given in the succeeding text: In a flame-dried, two-necked flask with septum caps and a stirring bar, n -BuPhZn (**1ab**) (10 mmol) was prepared using THF solution of ZnCl₂ (10 mmol), PhMgBr (10 mmol) in THF, n -BuMgBr (10 mmol) in THF as explained in the previous text. To the stirred mixture, internal standard nonane (1 mmol) and n -Bu₃P catalyst (0.1 mmol) and then PhCOCl **2** (1 mmol) were added. While carrying out the pseudo-first-order kinetic experiments, the volume of the reaction mixture was kept constant in order to keep the concentration of benzoyl chloride (**2**) constant. The mixture was quickly thermostated by immersing the flask in cooling bath at constant 0 °C temperature on a magnetic stirrer. Aliquots (8–12) were withdrawn from the mixture by syringe under nitrogen atmosphere at 15 s–2 min intervals and were hydrolyzed in a vial containing 1 N HCl solution. The aqueous phase was extracted with ether, and the ethereal phase was analyzed by gas chromatography (GC). The rate data were collected by measuring the concentration of coupling product **4** in each sample. Self consistent data could be obtained for two-four half lives. The uncertainty of the rate constants was lower than 12%, which is in the error limit of GC analysis, that is, generally ± 5 –14%.

RESULTS AND DISCUSSION

Our recent work showed that tri- n -butylphosphine-catalyzed acylation of n -BuPhZn (**1ab**) with benzoyl (**2**) chloride takes place with selective n -Bu group transfer (Scheme 2).



Scheme 2. Phosphine-catalyzed acylation reactions of diorganozinc *n*-BuRZn (**1**) with benzoyl chloride (**2**)



Scheme 3. Phosphine-catalyzed acylation of *n*-butylphenylzinc (**1ab**), di-*n*-butylzinc (**1a₂**), and *n*-butyl(substituted phenyl)zinc (**1ac-1ag**) reagents with benzoyl chloride (**2**) in tetrahydrofuran. These reactions are *n*-Bu group-selective

So, in the kinetic and mechanistic study on the phosphine-catalyzed acylation of alkyl(FG-C₆H₄)Zn reagents with benzoyl chloride aiming a new kinetic support for the dependence of reaction rate of transferable *n*-Bu group on the residual group, I planned to determine

- the rate constant of acylation of *n*-BuPhZn reagent (**1ab**),
- the rate constant of acylation of *n*-Bu₂Zn reagent (**1a₂**) to compare the transfer ability of *n*-Bu group in mixed and homo diorganozincs, and also
- the rate constant of acylation of *n*-Bu(FG-C₆H₄)Zn reagent (**1ac-1ag**) (FG=4-Me **c**, 3-Me **d**, 4-MeO **e**, 3-MeO **f**, 4-Br **g**) to evaluate the dependence of the transfer ability of *n*-Bu group on the residual FG-C₆H₄ group (Scheme 3).

For the evaluation of the rate data, I used the method that we already successfully used in the kinetics work of stoichiometric mixed cuprates, R¹R²CuMgBr^[56] and R¹R²CuZnI^[59] catalytic mixed cuprates,^[59,63] and also mixed diorganozincs, R¹R²Zn.^[65] In applying this method for concentration of electrophilic partner at time *t*, calculated value was used rather than directly measured value and it was calculated as $c = [\text{Electrophile}]_t = [\text{Electrophile}]_0 - \Sigma[\text{Product}]_t$. It was found that using this method gave more accurate results in the evaluation of the rate data of acylation reaction minimizing the errors due to the hydrolysis of the acylating agent during the work-up of aliquots.

As acylation of mixed diorganozincs *n*-Bu(FG-C₆H₄)Zn (**1ab-1ag**) give only **4** as product, $c_t = [\text{Benzoyl chloride}]_t$; that is, concentration of benzoyl chloride at time *t* was calculated as $c_t = [\text{Benzoyl chloride}]_0 - [n\text{-BuCOPh}]_t$. The rates of the reactions (i)–(iii) were studied by monitoring the formation of the product, *n*-BuCOPh (**4**) in THF at 0°C. The rate data were collected by taking at least 8–12 samples at 15 s–2 min intervals and doing the GC analysis of quenched aliquots to measure the concentration of **4**.

Kinetics of the acylation of *n*-BuPhZn and *n*-Bu₂Zn with benzoyl chloride in the presence of *n*-Bu₃P catalysis

A reasonable empirical rate law for the reaction (i) in Scheme 2 can be expressed in terms of concentrations of three components by Eqn 1.

$$\text{Rate} = -\frac{d[\text{PhCOCl}]}{dt} = k[n\text{-BuPhZn}]^x[\text{PhCOCl}]^y[n\text{-Bu}_3\text{P}]^z \quad (1)$$

Through a series rate data collected under pseudo-conditions, the reaction order in *n*-BuPhZn (**1ab**), PhCOCl (**2**), and *n*-Bu₃P (**3**) was found.

Determination of the order *x* in *n*-BuPhZn (**1ab**) and *y* in PhCOCl (**2**): at a constant and low catalyst *n*-Bu₃P (**3**) concentration, the concentration of *n*-BuPhZn (**1ab**) was used in excess to find the order *y* in PhCOCl (**2**). Rate can be written as

$$\text{Rate} = k_1[\text{PhCOCl}]^y \quad (2)$$

whereas pseudo-rate constant *k*₁ can be expressed using Eqn 3.

$$k_1 = k[n\text{-BuPhZn}]^x[n\text{-Bu}_3\text{P}]^z \quad (3)$$

As [*n*-Bu₃P] is kept constant, Eqn 3 can be also expressed as

$$k_1 = k_2[n\text{-BuPhZn}]^x \quad (4)$$

Pseudo-rate constant *k*₂ shows dependence on catalyst loading:

$$k_2 = k[n\text{-Bu}_3\text{P}]^z \quad (5)$$

The reaction order *x* in *n*-BuPhZn (**1ab**) was determined by changing the concentration of *n*-BuPhZn at the constant concentration of PhCOCl (**2**) and *n*-Bu₃P (**3**) in a series of reactions. According to Eqn 4, the slope of the log *k*₁ → log[*n*-BuPhZn] plot gives the value of *x*. For this purpose, the concentration of *n*-BuPhZn (**1ab**) was changed between 7 and 13 times than that of PhCOCl (**2**) while the concentration of *n*-Bu₃P (**3**) is 10 mol% of PhCOCl. The rate data were evaluated as pseudo-first order because log *c_t* values versus time proved linear up to 50–80% completion of the reaction. As an example for a pseudo-first-order plot, see Fig. S1a. Pseudo-first-order rate constants *k*₁ were calculated by linear regression analysis (*r*² > 0.98). For rate constants, *k*₁, see Table S1, and for the linearity of *k*₁ values with excess concentration of *n*-BuPhZn (**1ab**), see Fig. S2. The plot of log *k*₁ → log[*n*-BuPhZn] gave a slope of 1.08 (*r*² = 0.97) indicating the first-order reaction in *n*-BuPhZn. These results confirm the pseudo-second-order acylation of *n*-BuPhZn in the presence of phosphine catalysis. Pseudo-second-order rate constants *k*₂ calculated as *k*₂ = *k*₁/[*n*-BuPhZn] are also given in Table S1. Taking the average led to a rate constant of *k*₂ = 0.362 ± 0.013 M^{−1} min^{−1} within the uncertainty of max 4%, which is in the error limit of GC analysis.

The phosphine-catalyzed acylation of *n*-BuPhZn (**1ab**) under pseudo-conditions is quite fast and virtually complete within 8–12 min. So I carried out several trials to obtain reproducible results. Order in *n*-BuPhZn (**1ab**) was also established directly by determining the overall reaction order at equimolar concentration of **1ab** and benzoyl chloride (**2**) in the presence of catalyst *n*-Bu₃P (**3**). The plot of 1/*c_t* values versus time gave a straight line indicating that the reaction is second order in the presence of catalyst and therefore the reaction is first order in (**1ab**) and (**2**) (Fig. S1b). The rate constant was found to be *k*₂ = 0.34 M^{−1} min^{−1}. As seen, rate constants *k* calculated by applying pseudo-first-order or pseudo-second-order conditions resulted in the same values within the same error limit.

Determination of the order *z* in catalyst *n*-Bu₃P (**3**): As shown in Eqn 5, pseudo-second-order rate constant *k*₂ depends on phosphine catalyst (**3**) concentration; the reaction

order in catalyst was established by determining the rate constant k_2 at various catalyst concentrations. For this purpose, rate constants k_2 were determined at catalyst loadings of 10–50 mol% under pseudo-first-order conditions, that is, using excess n -BuPhZn (**1ab**) ($[n\text{-BuPhZn}]:[\text{PhCOCl}] = 7:1$). The results of this set of experiments are given in Table S2 and linear plot of k_2 values against concentration of n -Bu₃P (**3**) catalyst are given in Fig. S3. Subsequent plot of $\log k_2 \rightarrow \log [n\text{-Bu}_3\text{P}]$ yielded catalyst order of 1.23 ($r^2 = 0.98$). Using excess n -BuPhZn (**1ab**) in $[n\text{-BuPhZn}]:[\text{PhCOCl}] = 8:1$ ratio for rate determination in catalyst loading gave a similar catalyst order of 1. As orders close to 1 are most logically explained by unimolecular dependence, it is reasonable to conclude that the acylation reaction proceeds through a catalytic intermediate involving one molecule of n -Bu₃P (**3**).

The reaction rate constants k were calculated as $k = k_2/[n\text{-Bu}_3\text{P}]$ using k_2 values found applying pseudo-first-order benzoylation of n -BuPhZn (**1ab**) in the presence of n -Bu₃P (**3**) and they are also included in Table S2. Taking the average gave $k = 193.2 \pm 23.7 \text{ M}^{-2} \text{ min}^{-1}$ with the uncertainty of max, 12%, which is in the error limit of GC analysis.

Thus, the rate data for phosphine-catalyzed acylation of n -BuPhZn (**1ab**) can be described accurately by the rate law in Eqn 6.

$$\text{Rate} = -\frac{d[\text{PhCOCl}]}{dt} = k[n\text{-BuPhZn}][\text{PhCOCl}][n\text{-Bu}_3\text{P}] \quad (6)$$

I carried out a second set of experiments to find the rate constant of n -Bu₃P (**3**)-catalyzed acylation of n -Bu₂Zn (**1a₂**) with benzoyl chloride (**3**) in THF at 0 °C (Scheme 2(2)) to compare the reaction rates of mixed n -BuPhZn (**1ab**) and homo n -Bu₂Zn (**1a₂**) reagents. For this purpose, the rate constant k' was measured in similar pseudo-first-order conditions used for the rate data of **1ab**. First, the pseudo-first-rate constant k'_1 in the presence of excess **1a₂** and 10 mol% n -Bu₃P was determined. An example for a pseudo-first-order plot is given in Fig. S1c. k'_1 values are listed in Table S3, and linearity of rate constants k'_1 against excess concentration of n -Bu₂Zn (**1a₂**) is shown in Fig. S4. The plot of $\log k'_1 \rightarrow \log [n\text{-Bu}_2\text{Zn}]$ gave a slope of 1.01 ($r^2 = 0.98$) indicating the first-order reaction in **1a₂**. Following confirmation the pseudo-second-order acylation of **1a₂** in the presence of n -Bu₃P (**3**) rate constants, k'_2 were calculated as $k'_2 = k'_1/[n\text{-Bu}_2\text{Zn}]$ (Table S3) and taking the average gave $k'_2 = 2.46 \pm 0.05 \text{ M}^{-1} \text{ min}^{-1}$ with the uncertainty of 2% at 10 mol% n -Bu₃P. As n -Bu₃P-catalyzed benzoylation of n -BuPhZn (**1ab**) was found to be first order in n -Bu₃P, the determination of the pseudo-second-order rate constants in the presence of varying concentrations of n -Bu₃P was not repeated. The reaction rate constant was calculated as $k' = k'_2/[n\text{-Bu}_3\text{P}]$ and found to be $934.8 \text{ M}^{-2} \text{ min}^{-1}$ with the uncertainty of 2%.

The total rate law for the phosphine-catalyzed acylation of n -Bu₂Zn (**1a₂**) can be written as

$$\text{Rate} = -\frac{d[\text{PhCOCl}]}{dt} = k'[n\text{-Bu}_2\text{Zn}][\text{PhCOCl}][n\text{-Bu}_3\text{P}] \quad (7)$$

Effect of residual group FG–C₆H₄ on the transfer rate of n -Bu group in the acylation of n -Bu(FG–C₆H₄)Zn with benzoyl chloride in the presence of n -Bu₃P catalysis

Before providing an explanation for the rate laws expressed by Eqns 6 and 7 and proposing a reaction pathway for the acylation

of diorganozincs under phosphine catalysis, it also seemed quite interesting to test the dependence of transfer rate of n -Bu group to electronic effects of FG substituent on residual group in the phosphine-catalyzed acylation of n -Bu(FG–C₆H₄)Zn reagents. For this purpose, Erdik *et al.* already reported the correlation of n -Bu group transfer rate to electronic effects of residual FG–C₆H₄ group in the alkylation of n -Bu(FG–C₆H₄)CuMgBr reagents^[57] and acylation of n -Bu(FG–C₆H₄)CuZnI reagents.^[59] Hammett plots with positive slopes were in accordance with the proposed mechanism for these reactions of mixed cuprate. However, the reactivity of organozinc reagents is known to increase depending on the negative charge on the Zn center. This is in good agreement with the lower rate constant for n -Bu group transfer in mixed diorganozinc reagent n -BuPhZn (**1ab**) than that in homo diorganozinc reagent n -Bu₂Zn (**1a₂**) to the acyl electrophile. The possible explanation is simply that $-I$ effect of Ph group in n -BuPhZn (**1ab**) decreases the carbanionic charge leading to a less ionic n -Bu–Zn bond compared with that in n -Bu–Zn– n -Bu (**1a₂**). As seen, the reaction rate of transferable R_T group in mixed diorganozincs R_RR_TZn depends upon the residual group R_R. I simply expected that in the acylation of n -Bu group of n -Bu(FG–C₆H₄)Zn (**1ab–1ag**) reagents, n -Bu group transfer will be favored with $+I, +M$ type (alkyl) and $-I > +M$ type (alkoxy) substituents whereas it will be inhibited $-I \gg +M$ type (halogen) substituents resulting in a Hammett plot with a negative slope.

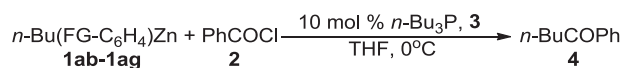
To test Hammett correlation, the rate constants were determined for the reaction of benzoyl chloride (**2**) with a small set of n -Bu(FG–C₆H₄)Zn reagents (**1ab–1ag**) (FG=H **b**, 4-Me **c**, 3-Me **d**, 4-MeO **e**, 3-MeO **f**, 4-Br **g**) under n -Bu₃P (**3**) catalysis (Scheme 2(3)). The second-order reaction rate constants k_2 were determined in pseudo-first-order conditions at constant n -Bu₃P concentration similar to that used for acylation of n -BuC₆H₅Zn (**1ab**) (Table 1), and the r^2 value was found close to unity for the slopes of each line in the kinetic experimental data of the reagents **1ab–1ag**.

It is pleasant that the resulting Hammett plot of rate constants k of n -Bu(FG–C₆H₄)Zn reagents (**1ab–1ag**) gave a negative slope, $\rho = -0.29$, with a good correlation ($r^2 = 0.910$) (Fig. 1).^[75] This result confirms Erdik's hypothesis on the dependence of transfer rate of R_T group to the electronic effect of residual R_R groups in R_RR_TCuM and R_RR_TZn reagents. This small value of reaction constant indicates a negligible effect of donation of electron density on R_R = FG–C₆H₄ group to R_T = n -Bu group via C–Zn–C bonds to react with C=O electrophile.

Reaction pathway for the phosphine-catalyzed acylation of diorganozincs

For the phosphine-catalyzed acylation of diorganozincs, a catalytic cycle has been already given^[61] depending on the formation of acyl n -butylphosphonium ions, RCOP[⊕](n -Bu)₃Cl[⊖] **A**.^[76–78]

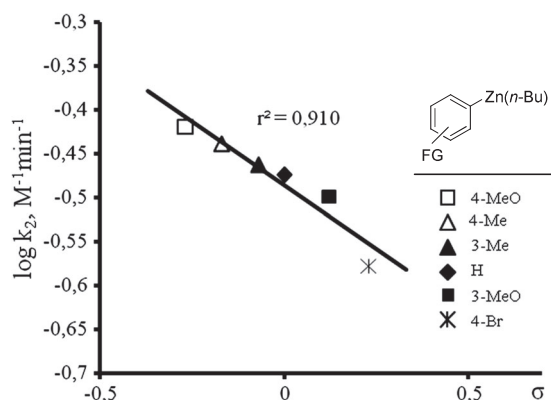
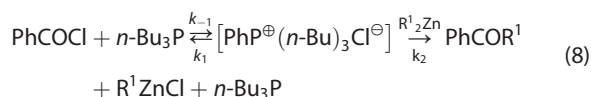
According to the kinetic data obtained in this study, a pathway for the catalytic acylation is given in Eqn 8. Three molecular collisions are rather rare in organic reactions,^[79,80] and in this case, the reaction does not take place in one step. The formation of the acylation product R¹COR **4** in the reaction of diorganozinc (**1**), acyl halide (**2**), and phosphine (**3**) could involve first formation of catalytic acylphosphonium complex **A** from **2** and **3** and then reaction of **A** with **1**. The mechanism consists of an equilibrium as a first step and the rate-determining step is either first step or second step depending on the observed third-order reaction kinetics.

Table 1. Rate constants k of acylation of n -butyl(substituted phenyl)zinc reagents (**1ab–ag**) with benzoyl chloride (**2**) in the presence of n -Bu₃P (**3**) in tetrahydrofuran at 0 °C^a

FG	$k_2, \text{M}^{-1} \text{min}^{-1}$
H, b ^b	0.336
4-Me, c	0.364
3-Me, d	0.344
4-MeO, e	0.381
3-MeO, f	0.317
4-Br, g	0.264

^a $[n\text{-Bu}(\text{FG-C}_6\text{H}_4)\text{Zn}] = 0.236 \text{ M}$, $[\text{PhCOCl}] = 0.026 \text{ M}$, $[n\text{-Bu}_3\text{P}] = 0.0026 \text{ M}$.

^bFrom Table S1.

**Figure 1.** Hammett plot for the n -Bu₃P (**3**)-catalyzed acylation of n -Bu(FG-C₆H₄)Zn reagents (**1ab–1ag**) with benzoyl chloride (**2**) in tetrahydrofuran at 0 °C

According to the steady-state kinetics, an explanation is provided for the rate laws expressed by Eqns 1 and 2. In the first step, catalytic intermediate **A** is formed and it is used up to form PhCOCl (**2**) and n -Bu₃P (**3**) again by the reverse reaction or to form the product **4** by reacting with R^1_2Zn (**1**) in the second step. As **A** is a reactive unstable species, it is expected that **A** is consumed at a rate at which it is formed:

$$k_1[\text{PhCOCl}][n\text{-Bu}_3\text{P}] = k_{-1}[\text{PhCOP}^\oplus(n\text{-Bu})_3\text{Cl}^\ominus] + k_2[\text{PhCOP}^\oplus(n\text{-Bu})_3\text{Cl}^\ominus][\text{R}^1_2\text{Zn}] \quad (9)$$

and rearrangement of Eqn 4 provides the expression in Eqn 10 for $[\text{PhCOP}^\oplus(n\text{-Bu})_3\text{Cl}^\ominus]$.

$$[\text{PhP}^\oplus(n\text{-Bu})_3\text{Cl}^\ominus] = -\frac{k_1[\text{PhCOCl}][n\text{-Bu}_3\text{P}]}{k_{-1} + k_2[\text{R}^1_2\text{Zn}]} \quad (10)$$

By using the value for $[\text{PhCOP}^\oplus(n\text{-Bu})_3\text{Cl}^\ominus]$, the following rate expression can be obtained for the second step:

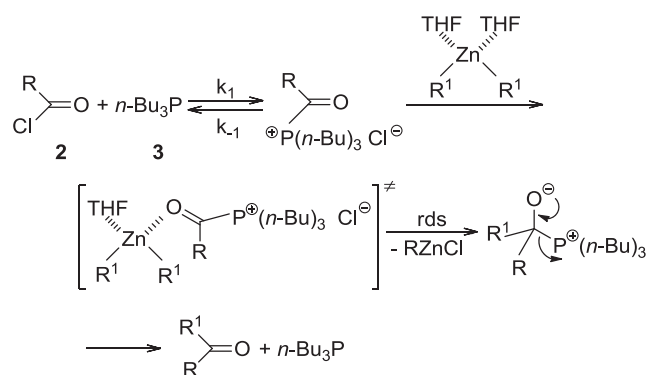
$$\begin{aligned} \text{Rate} &= k_2[\text{PhCOP}(n\text{-Bu})_3\text{Cl}][\text{R}^1_2\text{Zn}] \\ &= \frac{k_1k_2[\text{PhCOCl}][n\text{-Bu}_3\text{P}][\text{R}^1_2\text{Zn}]}{k_{-1} + k_2[\text{R}^1_2\text{Zn}]} \end{aligned} \quad (11)$$

If the first step is considered as a rapid pre-equilibrium and the second step is a rate-determining step, that is, $k_{-1} \gg k_2[\text{R}^1_2\text{Zn}]$, the following expression is obtained:

$$\text{Rate} = \frac{k_1k_2}{k_{-1}}[\text{PhCOCl}][n\text{-Bu}_3\text{P}][\text{R}^1_2\text{Zn}] \quad (12)$$

As seen, the simplified rate law corresponds to the rate equation for the catalyzed acylation, which is first order in diorganozinc (**1**), acyl halide (**2**), and phosphine catalyst (**3**) as observed.

For acylation of diorganozincs (**1**) with catalytic acylphosphonium ion **A**, an addition–elimination mechanism is also given in Scheme 4, with a transition state similar to that proposed for the addition of diorganozincs to aldehydes.^[81–83]

**Scheme 4.** Proposed mechanism for the phosphine-catalyzed acylation of diorganozinc reagents

Proposed reaction mechanism for the phosphine-catalyzed acylation is supported by the sign and magnitude of ρ in the Hammett plot. Negative reaction constant ρ indicates that the reaction of diorganozincs **1** with acylphosphonium ion **A** corresponds to the rate-determining step, that is, the second step in reaction pathway (Eqn 3). A small interaction of FG on the FG-C₆H₄ group with the reaction center n -Bu in n -Bu(FG-C₆H₄)Zn reagent in the transition state is consistent with the low value of ρ .

CONCLUSION

A detailed kinetic study of the phosphine-catalyzed acylation of mixed diorganozinc, n -BuPhZn, homo diorganozinc, n -Bu₂Zn, and n -Bu(FG-C₆H₄)Zn reagents was reported with the aim of gaining support for the hypothesis that the rate of transferable group R_T in $\text{R}_\text{R}\text{R}_\text{T}\text{Zn}$ reagents on the residual group R_R . The kinetic law for the reaction in THF at 0 °C was found to be $\text{rate} = k[\text{R}_2\text{Zn}][\text{acyl halide}][n\text{-Bu}_3\text{P}]$. Hammett methodology was also used in the acylation of n -Bu(FG-C₆H₄)Zn reagents for the correlation of the reaction rate of transferable group

n-Bu to the electronic effects of residual FG-C₆H₄ group. The kinetic data were offered. The lower reaction rate of *n*-BuPhZn than that of *n*-Bu₂Zn and negative reaction constant in Hammett plot shows electron-withdrawing effect of residual groups on the carbanionic charge of transferable group *n*-Bu in the acylation reaction. These findings support the hypothesis and indicate that phosphine-catalyzed acylation of diorganozincs takes place through the formation of acylphosphonium salt.

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^sCorrection added on 2 February 2016, after first online publication: the first author's name in reference 72 was corrected.